MS&E 228: Applied Causal Inference Powered by ML and AI

Lecture 19: Surrogates for Long-Term Treatment Effects

Vasilis Syrgkanis

Stanford University

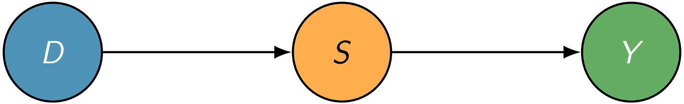
Winter 2026

Estimating Long-Term Returns on Investment

- Companies frequently deploys new discount or customer support programs
- Which of these programs ("investments") are more successful than others?
- Success is a long-term objective: what is the effect of the program on the two-year customer journey (e.g., effect on two-year revenue)
- We cannot wait two years to evaluate a program
- Main Question. Can we construct estimates of the values of these programs with short-term data, e.g. after 6 months?

Long-Term Effects from Short-Term Surrogates

- Suppose that there are many short-term signals S that are indicative of a customer's long-term reward Y (e.g. the next 6-month purchase patterns of a customer could be indicative of their long-term spend)
- Suppose that investment program D affects long-term rewards if and only if it affects these short-term signals



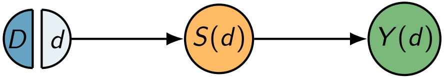
- We will call these short-term signals S surrogates

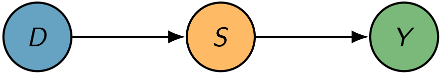
Causal Inference with Surrogates 101

Since long-term effect goes only through surrogates:

$$(\textit{effect on long-term reward}) = (\textit{effect on forecasted long-term reward based on surrogates})$$

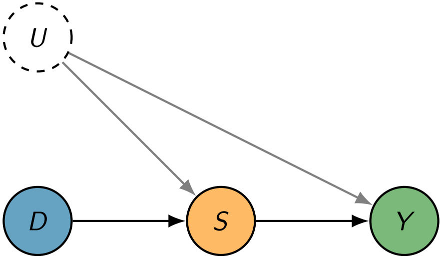
$$
\begin{aligned}
\mathbb{E}[Y(d)] &= \mathbb{E}[Y|D=d] && \text{(Exogeneity of Treatment)}\\
&= \mathbb{E}[\mathbb{E}[Y|D,S]|D=d] && \text{(Tower Law)}\\
&= \mathbb{E}[\underbrace{\mathbb{E}[Y|S]}_{\substack{d\text{-separation of } D \text{ from } Y\\ \text{by } S \text{ in DAG}}}|D=d] = \mathbb{E}[\underbrace{h(S)}_{\substack{\text{Forecasted Outcome}\\ \text{from surrogates}}}|D=d]
\end{aligned}
$$





Caution: Surrogates should not be confounded!

Surrogates are confounded with an unobserved confounder:



$$(\textit{effect on long-term reward}) \neq (\textit{effect on forecasted long-term reward based on surrogates})$$

The key property that we used in the previous slide does not hold:

$$Y \perp\!\!\!\perp D \mid S$$

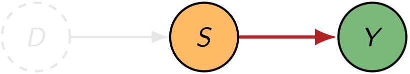
Collider path opens by conditioning! We can block it by conditioning on U (had we observed it)

$$Y \perp\!\!\!\perp D \mid S, U$$

Causal Inference with Surrogates 101

historical/long-term (ℓ)

- Contains long-term Y and surrogates S
- Relationship between Y and S the same as in short-term dataset (E)



$h(S) = E_\ell[Y|S]$

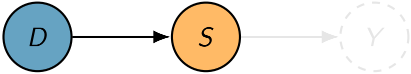
1. Estimate

$$h(S) := \mathbb{E}_\ell[Y|S] \quad \text{(surrogate index)}$$

from (ℓ) by regressing $Y \sim S$

recent/short-term (s)

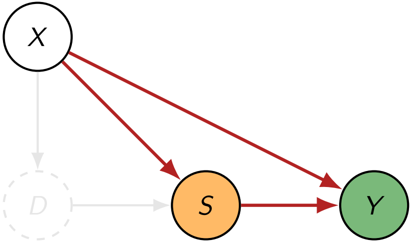
- Contains treatment D and surrogates S
- Does not contain long term outcome Y



2. Impute expected long-term outcomes in (s)
3. Regress $h(S) \sim D$ to estimate effect of D on long-term outcome Y from (E)

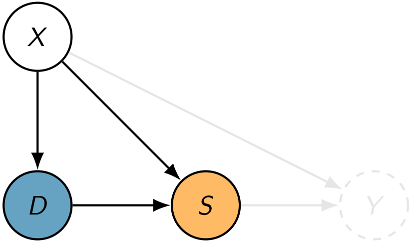
Causal Inference with Surrogates

long-term data (ℓ)



$h(S, X) = E_\ell[Y|S, X]$

short-term data (s)



1. Estimate

$$h(S, X) := \mathbb{E}_\ell[Y|S, X] \quad \text{(surrogate index)}$$

by regressing $Y \sim S, X$

2. Impute expected long-term outcomes in (s)
3. Estimate average effect of D on $h(S, X)$ by conditioning on X

Causal Inference with Surrogates 102

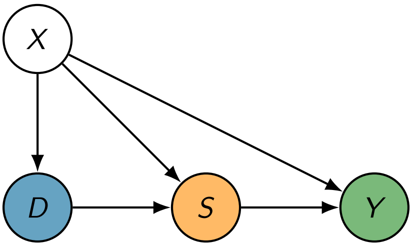
$$\tau(d) := \mathbb{E}_s[Y(d)] = \mathbb{E}_s[\mathbb{E}_s[Y|D = d, X]], \qquad (g\text{-formula}, \; Y(d) \perp\!\!\!\perp D \mid X)$$

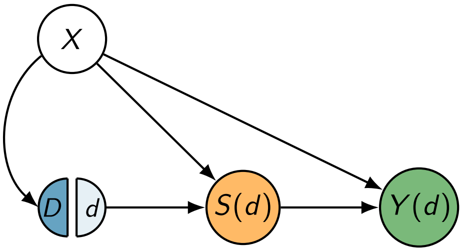
- Surrogacy assumption implies $Y \perp\!\!\!\perp D \mid S, X$:

$$\mathbb{E}_s[Y|D = d, X] = \mathbb{E}_s\left[\mathbb{E}_s[Y|D, S, X] \mid D = d, X\right] = \mathbb{E}_s\left[\mathbb{E}_s[Y \mid S, X] \mid D = d, X\right]$$

- Invariance between short and long term implies

$$\mathbb{E}_s[Y \mid S, X] = \mathbb{E}_\ell[Y \mid S, X] = h(S, X)$$





Nested Regression Estimand

Surrogacy based g-formula

$$\tau(d) = \mathbb{E}_s[g(d, X)]$$
$$g(D, X) = \mathbb{E}_s[h(S, X)|D, X]$$
$$h(S, X) = \mathbb{E}_\ell[Y|S, X]$$

- Target parameter is averaged over short-term data distribution
- g regression function is trained on the outputs of the h regression function
- Surrogate Index Model h is trained on long-term data distribution

Debiased Insensitive Formula

Surrogacy based g-formula

$$\begin{aligned} \tau(d) &= \mathbb{E}_s[g(d, X)] \\ g(D, X) &= \mathbb{E}_s[h(S, X) | D, X] \\ h(S, X) &= \mathbb{E}_\ell[Y | S, X] \end{aligned}$$

Debiased formula

$$\tau(d) = E_s \left[g(d, X) + \underbrace{\frac{\mathbf{1}\{D = d\}}{\Pr_s(D = d \mid X)}}_{\text{Inverse Propensity Weights}} \cdot \underbrace{(h(S, X) - g(D, X))}_{g \text{ regression residual}} \right]$$

Recursive Debiasing

$$\mathbb{E}_s\left[g(d,X) + \underbrace{\frac{\mathbf{1}\{D=d\}}{\Pr_s(D=d|X)}}_{a(D,X)}\left(h(S,X) - g(D,X)\right)\right] + \mathbb{E}_\ell\left[w(S,X)\cdot \underbrace{(Y - h(S,X))}_{h \text{ regression residual}}\right]$$

The weighting function $w(S,X)$ is of the form

$$w(S,X) = s(S,X)\cdot \mathbb{E}_s[a(D,X)|S,X] = s(S,X)\cdot \frac{\Pr_s(D=d \mid S,X)}{\Pr_s(D=d \mid X)}$$

Surrogate Score s: density ratio of S,X between ℓ and s "environments"

$$s(S,X) = \frac{p(S,X|E=s)}{p(S,X|E=\ell)} = \frac{\Pr(E=s|S,X)\,\Pr(E=\ell)}{\Pr(E=\ell|S,X)\,\Pr(E=s)}$$

Doubly Robust Formula for ATE with Surrogates

For the ATE we just take the difference of the two formulas for $d = 1$ and $d = 0$:

$$\text{ATE} = \tau(1) - \tau(0)$$

Which simplifies to:

$$\theta = \mathbb{E}_s \left[g(1, X) + g(0, X) + \frac{D - p(X)}{p(X)\,(1 - p(X))} \left(h(S, X) - g(D, X) \right) \right]$$

$$+ \mathbb{E}_\ell \left[s(S, X) \left(\frac{\pi(S, X)}{p(X)} - \frac{1 - \pi(S, X)}{1 - p(X)} \right) \cdot (Y - h(S, X)) \right]$$

where

$$p(X) = \Pr_s(D = 1 \mid X) \qquad \pi(S, X) = \Pr_s(D = 1 \mid S, X)$$

DR Surrogate ATE Estimator: Pseudocode — Nuisance Estimation

```
def dr_surrogate_ate(data_s, data_l, S_cols, X_cols, K=5,  # ATE = E[Y(1)]-E[Y(0)]
                     reg_model, clf_model):
    SX, DX = S_cols + X_cols, ['D'] + X_cols
    n_s, n_l = len(data_s), len(data_l)
    h_hat_l, h_hat_s = np.zeros(n_l), np.zeros(n_s)
    g_hat, g1_hat, g0_hat = np.zeros(n_s), np.zeros(n_s), np.zeros(n_s)
    pi_hat, w1_hat, w0_hat = np.zeros(n_s), np.zeros(n_l), np.zeros(n_l)
    data_s['E'], data_l['E'] = 1, 0
    data_all = pd.concat([data_l, data_s])  # first n_l long-term, then n_s short-term
    for train, test in KFold(n_splits=K, shuffle=True).split(data_all):
        tr_l, te_l = train[train < n_l], test[test < n_l]
        tr_s, te_s = train[train >= n_l] - n_l, test[test >= n_l] - n_l
        # h=E_l[Y|S,X]
        h_m = clone(reg_model).fit(data_l.iloc[tr_l][SX], data_l.iloc[tr_l]['Y'])
        h_hat_l[te_l] = h_m.predict(data_l.iloc[te_l][SX])
        h_hat_s[te_s] = h_m.predict(data_s.iloc[te_s][SX])
        # g=E_s[h|D,X]
        g_m = clone(reg_model).fit(data_s.iloc[tr_s][DX], h_m.predict(data_s.iloc[tr_s][SX]))
        g_hat[te_s] = g_m.predict(data_s.iloc[te_s][DX])
        X_te = data_s.iloc[te_s][X_cols]
        te_1 = X_te.copy(); te_1.insert(0,'D',1); te_0 = X_te.copy(); te_0.insert(0,'D',0)
        g1_hat[te_s], g0_hat[te_s] = g_m.predict(te_1), g_m.predict(te_0)
        # Pr(D|X)
        pi_m = clone(clf_model).fit(data_s.iloc[tr_s][X_cols], data_s.iloc[tr_s]['D'])
        pi_hat[te_s] = pi_m.predict_proba(data_s.iloc[te_s][X_cols])[:, 1]
        # density ratio s(S,X)
        idx_all = np.r_[tr_l, tr_s + n_l]
        s_m = clone(clf_model).fit(data_all.iloc[idx_all][SX], data_all.iloc[idx_all]['E'])
        s_p = s_m.predict_proba(data_l.iloc[te_l][SX])
        s_ratio = (s_p[:,1] / s_p[:,0]) * (n_l / n_s)
        # Pr(D|S,X)
        a_m = clone(clf_model).fit(data_s.iloc[tr_s][SX], data_s.iloc[tr_s]['D'])
        a_p = a_m.predict_proba(data_l.iloc[te_l][SX])
        pi_l = pi_m.predict_proba(data_l.iloc[te_l][X_cols])[:, 1]
        w1_hat[te_l] = s_ratio * a_p[:,1] / pi_l          # w_1(S,X)
        w0_hat[te_l] = s_ratio * a_p[:,0] / (1 - pi_l)    # w_0(S,X)
```

DR Surrogate ATE Estimator: Pseudocode — Estimation & Inference

```
# (continued) Difference the two DR formulas and simplify:
# tau = E_s[g(1,X)-g(0,X)] + E_s[ipw*(h(S,X)-g(D,X))] + E_l[(w1-w0)*(Y-h(S,X))]
# where ipw = (D - pi(X)) / (pi(X)*(1-pi(X)))  [simplification of 1{D=1}/pi - 1{D=0}/(1-pi)]
D_s, Y_l = data_s['D'].values, data_l['Y'].values
ipw = (D_s - pi_hat) / (pi_hat * (1 - pi_hat))  # simplified IPW for ATE
term1 = np.mean(g1_hat - g0_hat)                  # plug-in: E[g(1,X) - g(0,X)]
term2 = np.mean(ipw * (h_hat_s - g_hat))           # debiasing for g
term3 = np.mean((w1_hat - w0_hat) * (Y_l - h_hat_l))  # debiasing for h
tau = term1 + term2 + term3
# Influence function for ATE
psi_s = (g1_hat - g0_hat) + ipw * (h_hat_s - g_hat) - tau  # n_s-vector
psi_l = (w1_hat - w0_hat) * (Y_l - h_hat_l)                # n_l-vector
# Asymptotic variance: Var(tau) = Var(psi_s)/n_s + Var(psi_l)/n_l
se = np.sqrt(np.var(psi_s) / n_s + np.var(psi_l) / n_l)
return tau, se, tau - 1.96 * se, tau + 1.96 * se
```

References I

Athey, S., Chetty, R., Imbens, G. W., & Kang, H. (2019).
The Surrogate Index: Combining Short-Term Proxies to Estimate Long-Term Treatment Effects More Rapidly and Precisely.
NBER Working Paper, (w26463).

Athey, S., Chetty, R., Imbens, G. W., & Kang, H. (2025).
The surrogate index: Combining short-term proxies to estimate long-term treatment effects more rapidly and precisely.
Review of Economic Studies, rdaf087.

Battocchi, K., Dillon, E., Hei, M., Lewis, G., Oprescu, M., & Syrgkanis, V. (2021).
Estimating the long-term effects of novel treatments.
Advances in Neural Information Processing Systems, 34, 2925-2935.

Bibaut, A., Kallus, N., Ejdemyr, S., & Zhao, M. (2023).
Long-term causal inference with imperfect surrogates using many weak experiments, proxies, and cross-fold moments.
arXiv preprint arXiv:2311.04657.

References II

Chen, J., & Ritzwoller, D. M. (2023).
Semiparametric estimation of long-term treatment effects.
Journal of Econometrics, 237(2), 105545.

Chernozhukov, V., Newey, W., Singh, R., & Syrgkanis, V. (2022).
Automatic Debiased Machine Learning for Dynamic Treatment Effects and General Nested Functionals.
arXiv preprint arXiv:2203.13887.

Prentice, R. L. (1989).
Surrogate endpoints in clinical trials: definition and operational criteria.
Statistics in medicine, 8(4), 431-440.

Imbens, G., Kallus, N., Mao, X., & Wang, Y. (2025).
Long-term causal inference under persistent confounding via data combination.
Journal of the Royal Statistical Society Series B: Statistical Methodology, 87(2), 362-388.